

MH371 broad-flight truth-separated control — uncertainty diagnostic

Broad multi-mode, repeated-event filtering diagnostic

Numerical support: FAILED

Held-back truth loaded for accuracy scoring

Scientific families (never pooled)

- BTO only: 8 deterministic seed(s)
- BTO + BFO (7 Hz): 8 deterministic seed(s)

Declared control-history support

Five initial modes (declared prior):
CTH 20.0% CMH 20.0% CTT 20.0%
CMT 20.0% GC 20.0%

Four recurring-event rates (declared prior):
tau=640s 25.0% tau=2024s 25.0%
tau=6402s 25.0% tau=20243s 25.0%

Every rate family enables recurring lateral, speed, and altitude renewal streams. Each lateral renewal can reselect any of the five modes and draw across the declared ± 180 -degree course-change support; speed and altitude commands renew independently. All displayed prior weights are declared and uncalibrated.

Interpretation limits

- No assumption-free or empirically calibrated true flight-path probability is claimed.
- Control modes, event-rate families, event marks, performance bounds, and their prior probabilities are declared model choices.
- Top-level scientific families are not pooled; seed pooling is numerical only and equal within a family.
- One known-flight trajectory cannot calibrate general coverage of the accident-flight technique.
- Map density is display-smoothed; all metrics use unsmoothed particle weights.

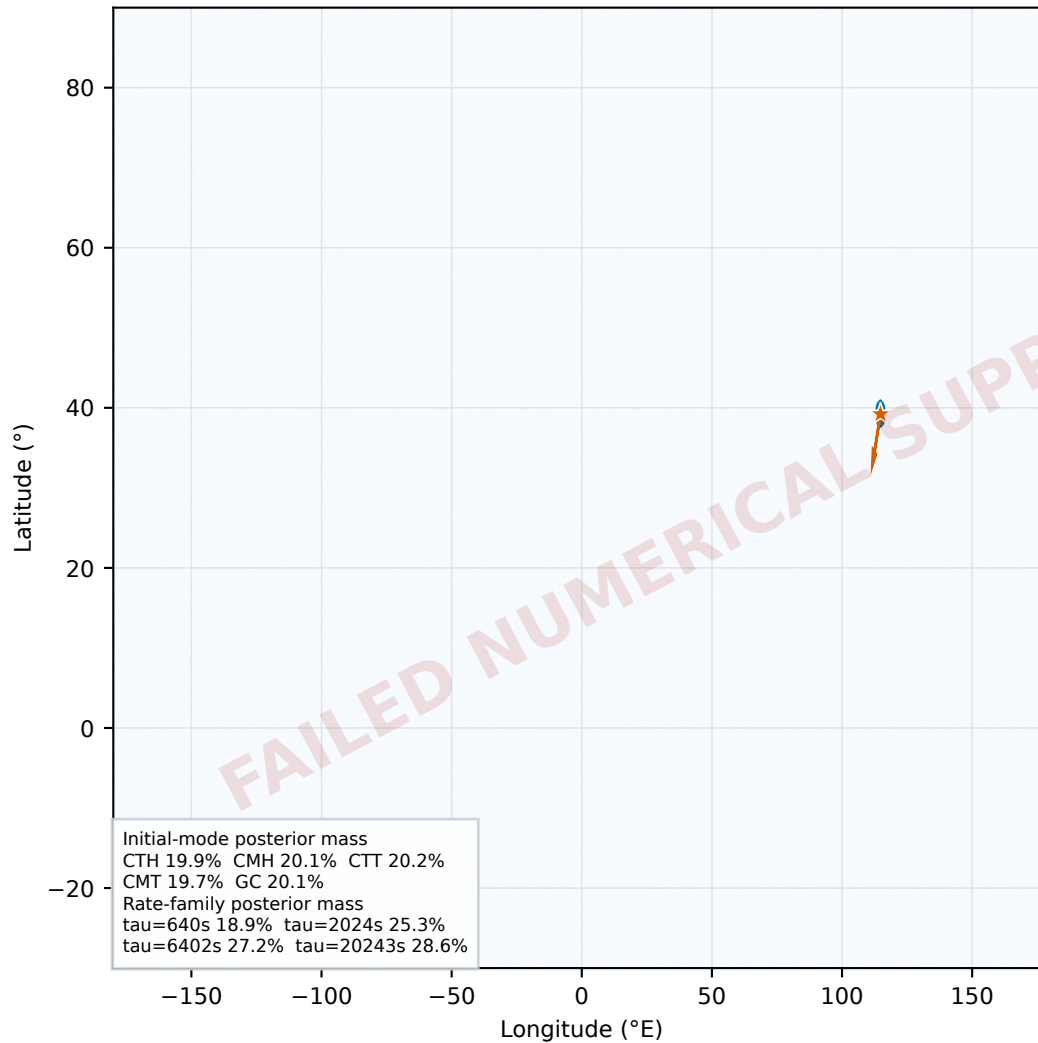
Failed/unassessed criteria: 1758 of 2611

Complete per-epoch criterion ledger: companion JSON audit.

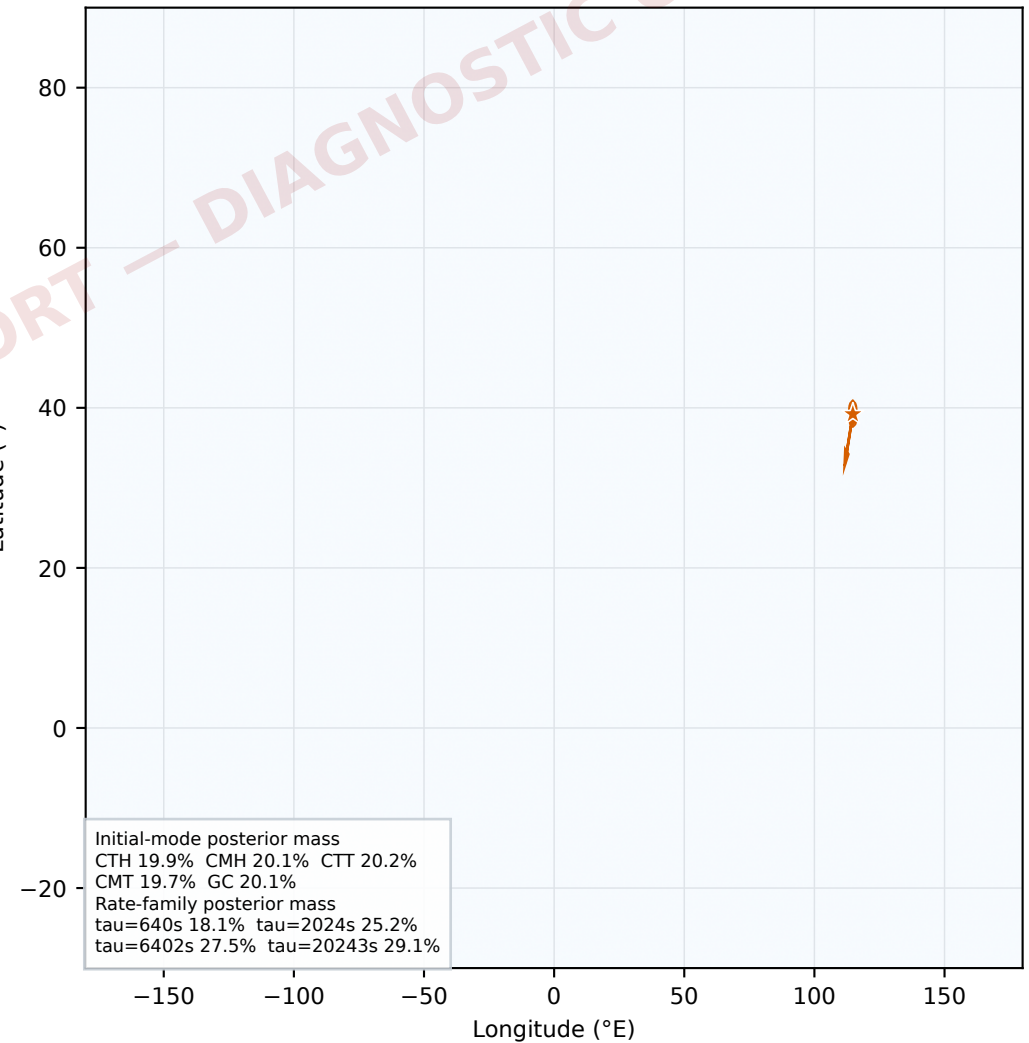
Every spatial probability shown is conditional on the declared model and evidence family. It is not a calibrated 'true uncertainty'.

MH371 — mh371_0200 filtering posterior

BTO only (kept separate)
SATCOM likelihood update: BTO
particle/root ESS 46599 / 46599.2
mean lateral/speed/altitude events 0.12 / 0.22 / 0.22
mass with ≥ 2 lateral events 1.2%
seed TV max 0.004
held-back mean error 9.9 NM
mass within 100 NM 100.0%

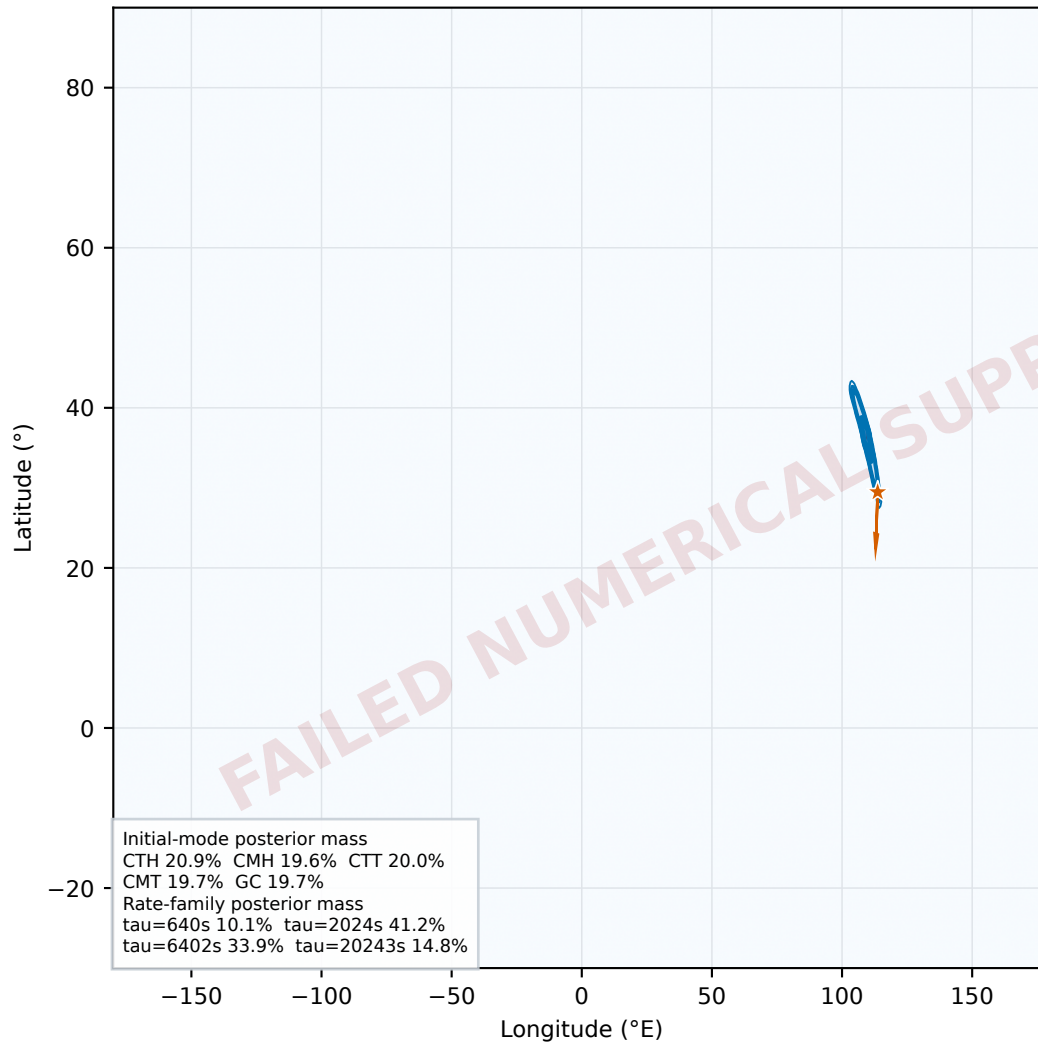


BTO + BFO (7 Hz) (kept separate)
SATCOM likelihood update: BTO + BFO
particle/root ESS 46016 / 46015.7
mean lateral/speed/altitude events 0.12 / 0.21 / 0.20
mass with ≥ 2 lateral events 1.1%
seed TV max 0.004
held-back mean error 9.8 NM
mass within 100 NM 100.0%

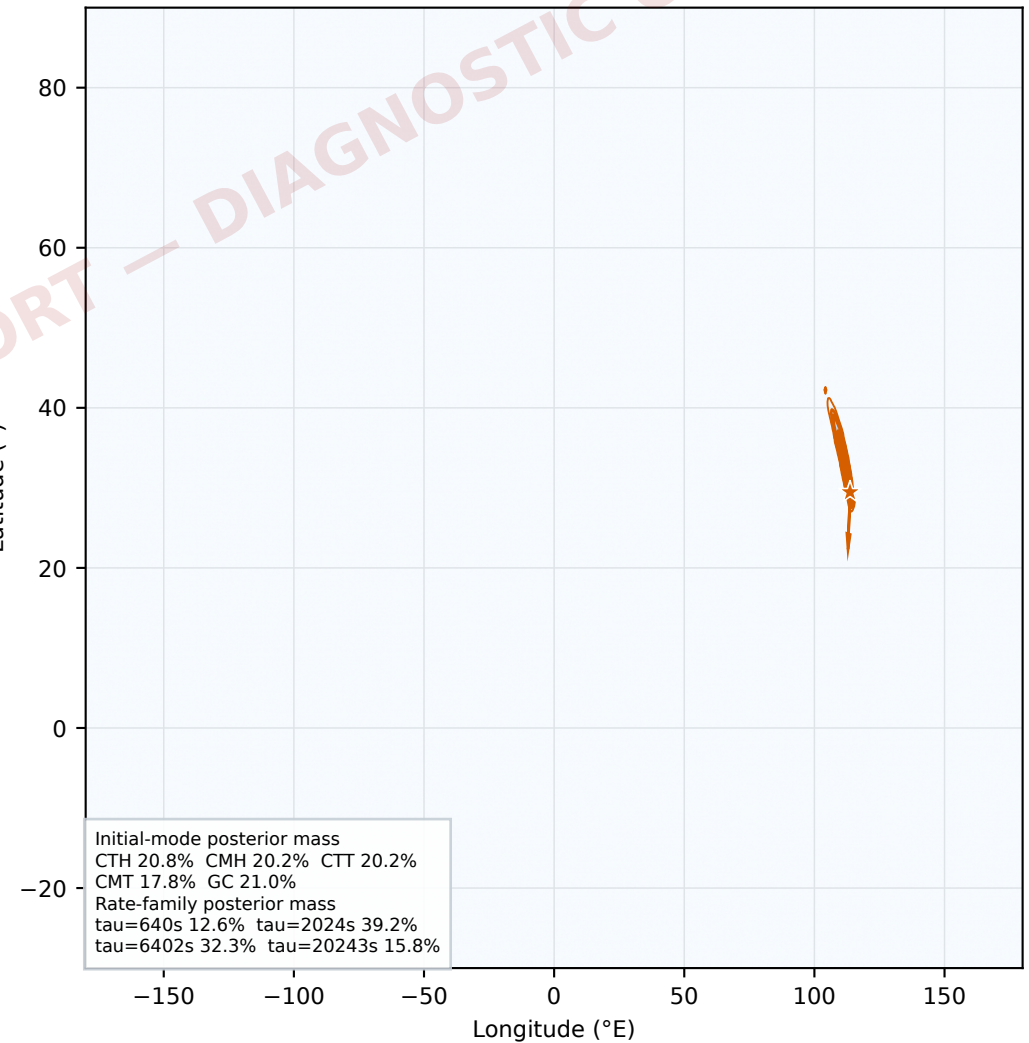


MH371 — mh371_0300 filtering posterior

BTO only (kept separate)
SATCOM likelihood update: BTO
particle/root ESS 1688 / 1424.8
mean lateral/speed/altitude events 2.21 / 2.47 / 2.27
mass with ≥ 2 lateral events 50.4%
seed TV max 0.277
held-back mean error 463.3 NM
mass within 100 NM 3.8%

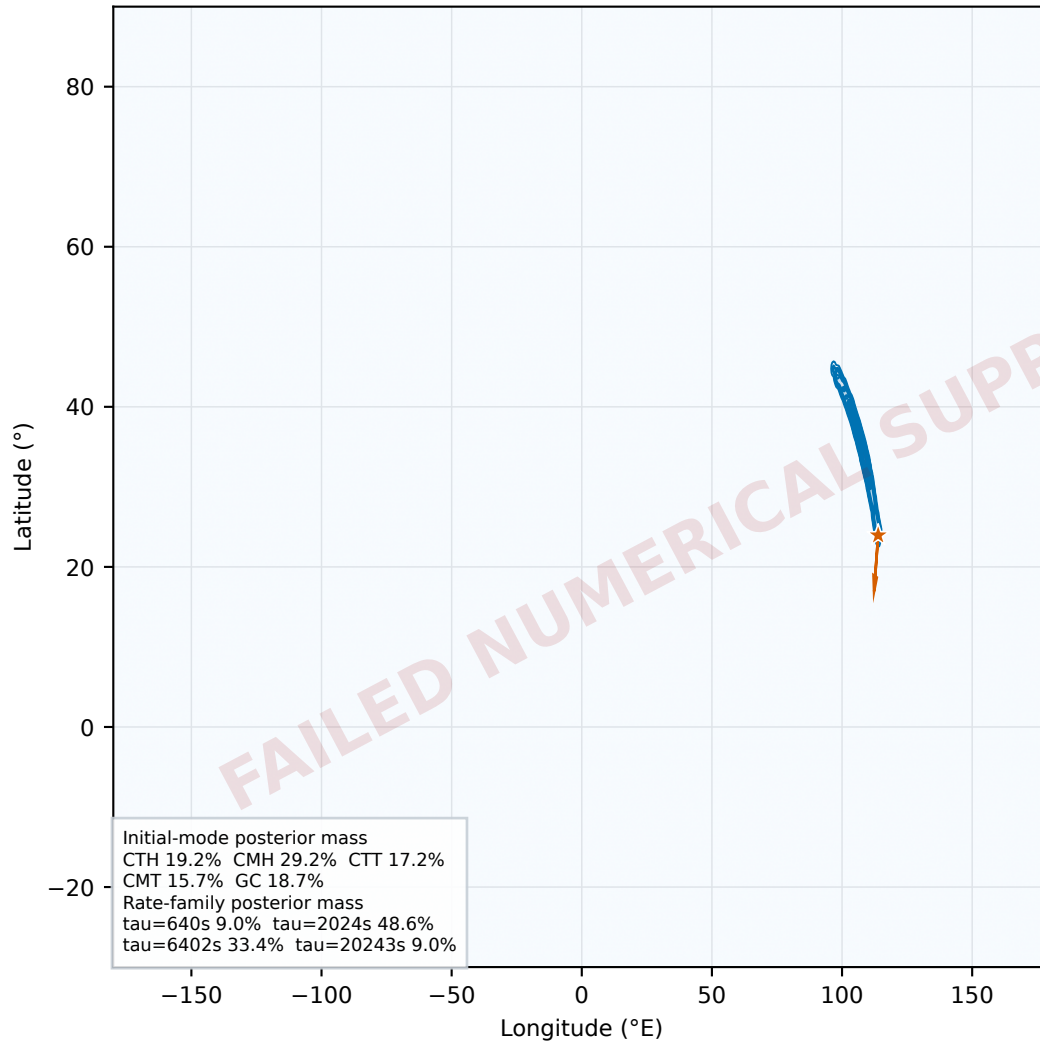


BTO + BFO (7 Hz) (kept separate)
SATCOM likelihood update: BTO + BFO
particle/root ESS 558 / 462.1
mean lateral/speed/altitude events 2.38 / 2.55 / 2.52
mass with ≥ 2 lateral events 52.1%
seed TV max 0.434
held-back mean error 292.5 NM
mass within 100 NM 14.1%

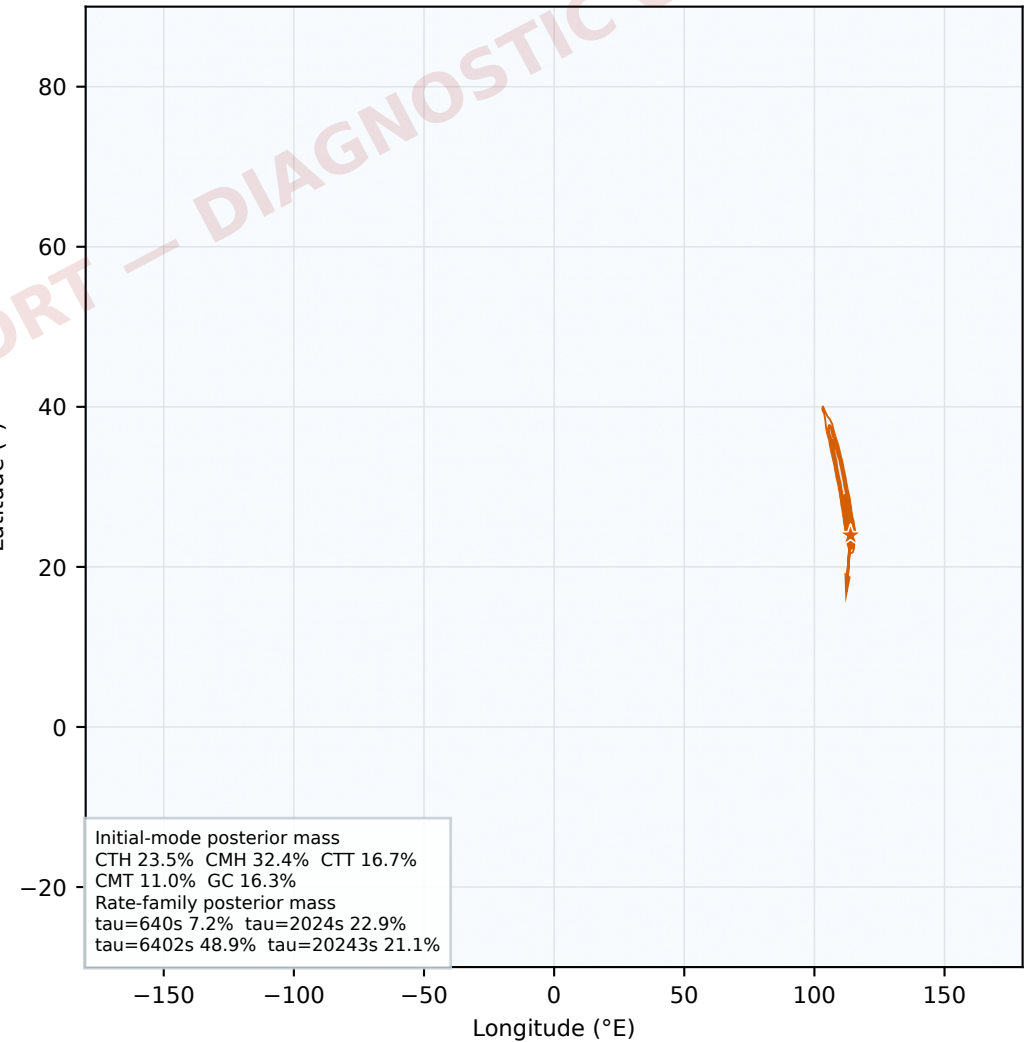


MH371 — mh371_0400 filtering posterior

BTO only (kept separate)
SATCOM likelihood update: BTO
particle/root ESS 2016 / 200.2
mean lateral/speed/altitude events 3.47 / 3.76 / 3.49
mass with ≥ 2 lateral events 70.9%
seed TV max 0.682
held-back mean error 748.9 NM
mass within 100 NM 6.8%

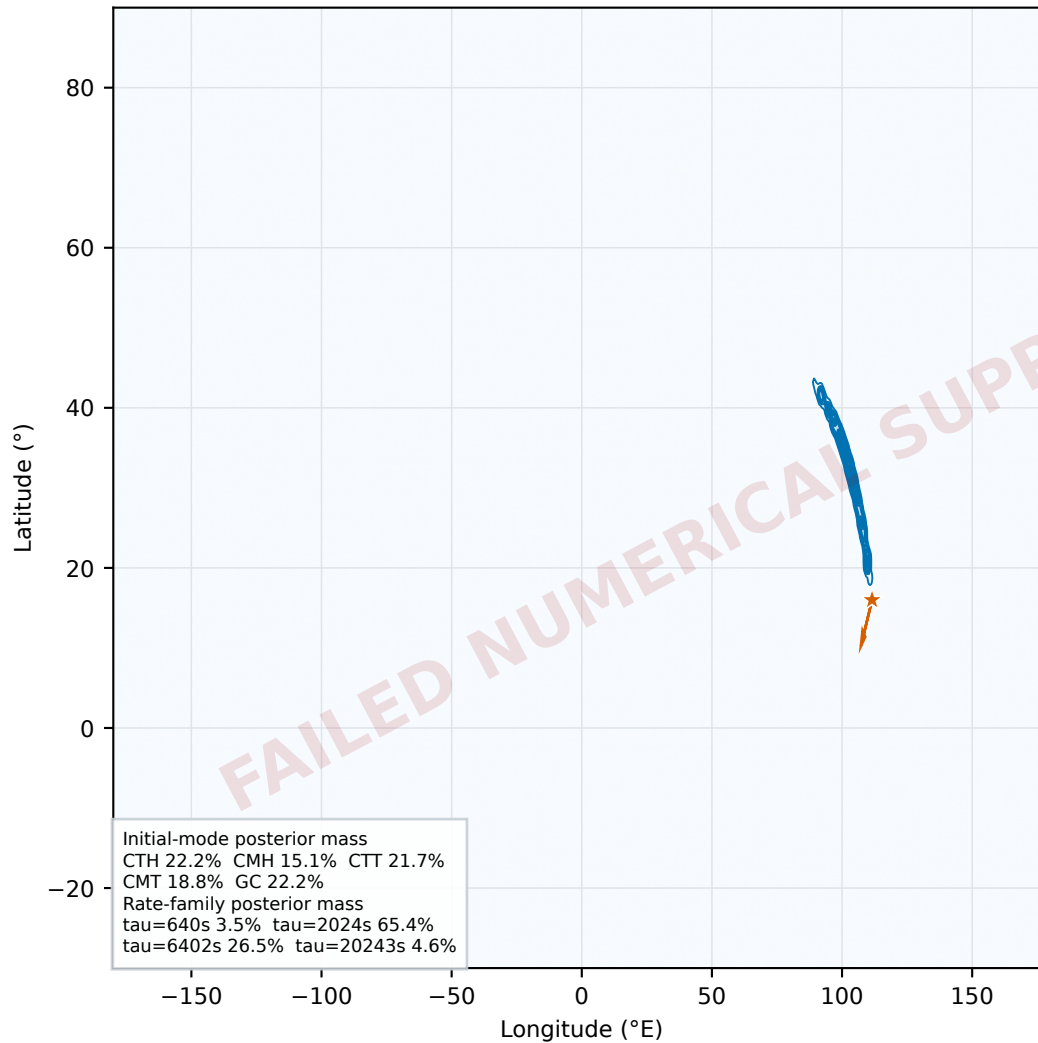


BTO + BFO (7 Hz) (kept separate)
SATCOM likelihood update: BTO + BFO
particle/root ESS 1078 / 26.1
mean lateral/speed/altitude events 2.56 / 2.39 / 2.46
mass with ≥ 2 lateral events 47.5%
seed TV max 0.954
held-back mean error 247.5 NM
mass within 100 NM 39.4%

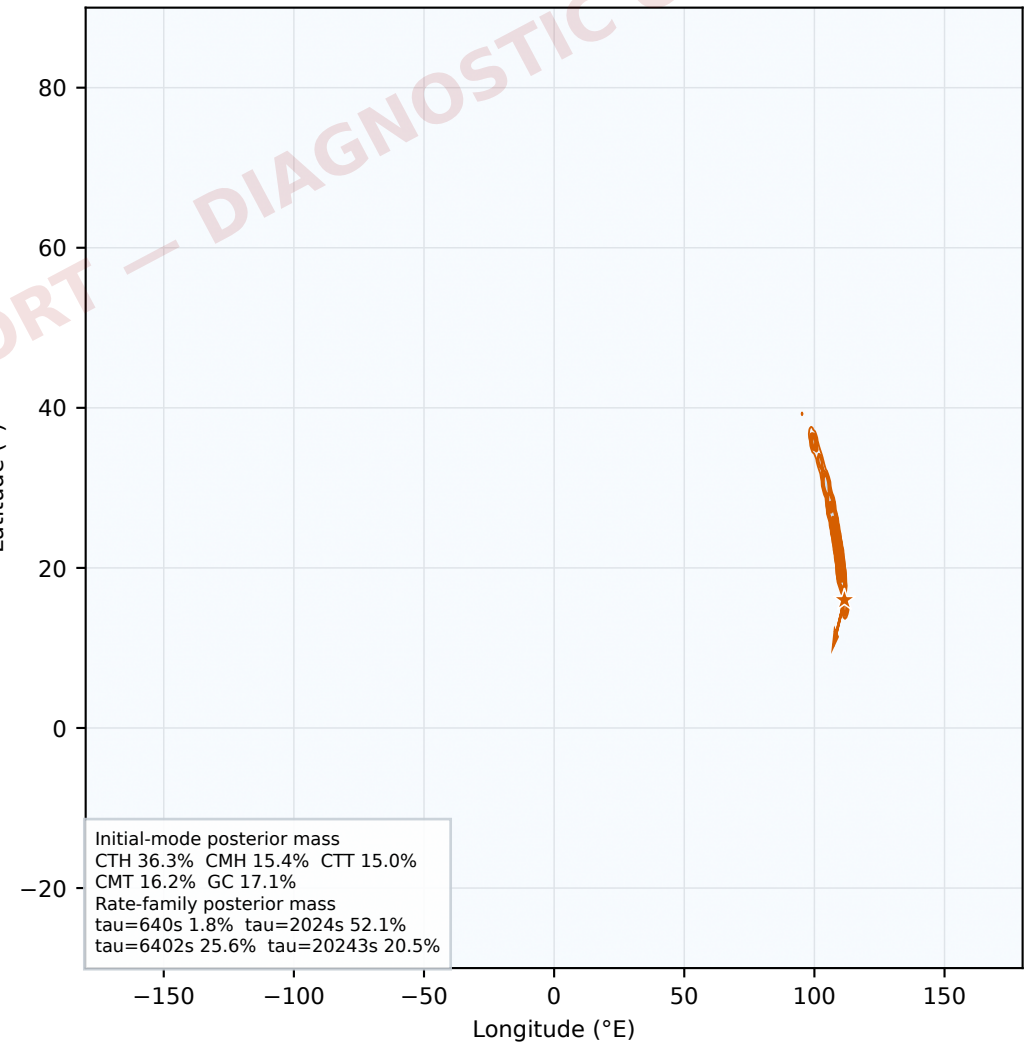


MH371 — mh371_0500 filtering posterior

BTO only (kept separate)
SATCOM likelihood update: BTO
particle/root ESS 466 / 88.5
mean lateral/speed/altitude events 4.80 / 5.46 / 4.93
mass with ≥ 2 lateral events 96.0%
seed TV max 0.799
held-back mean error 1048.7 NM
mass within 100 NM 0.1%

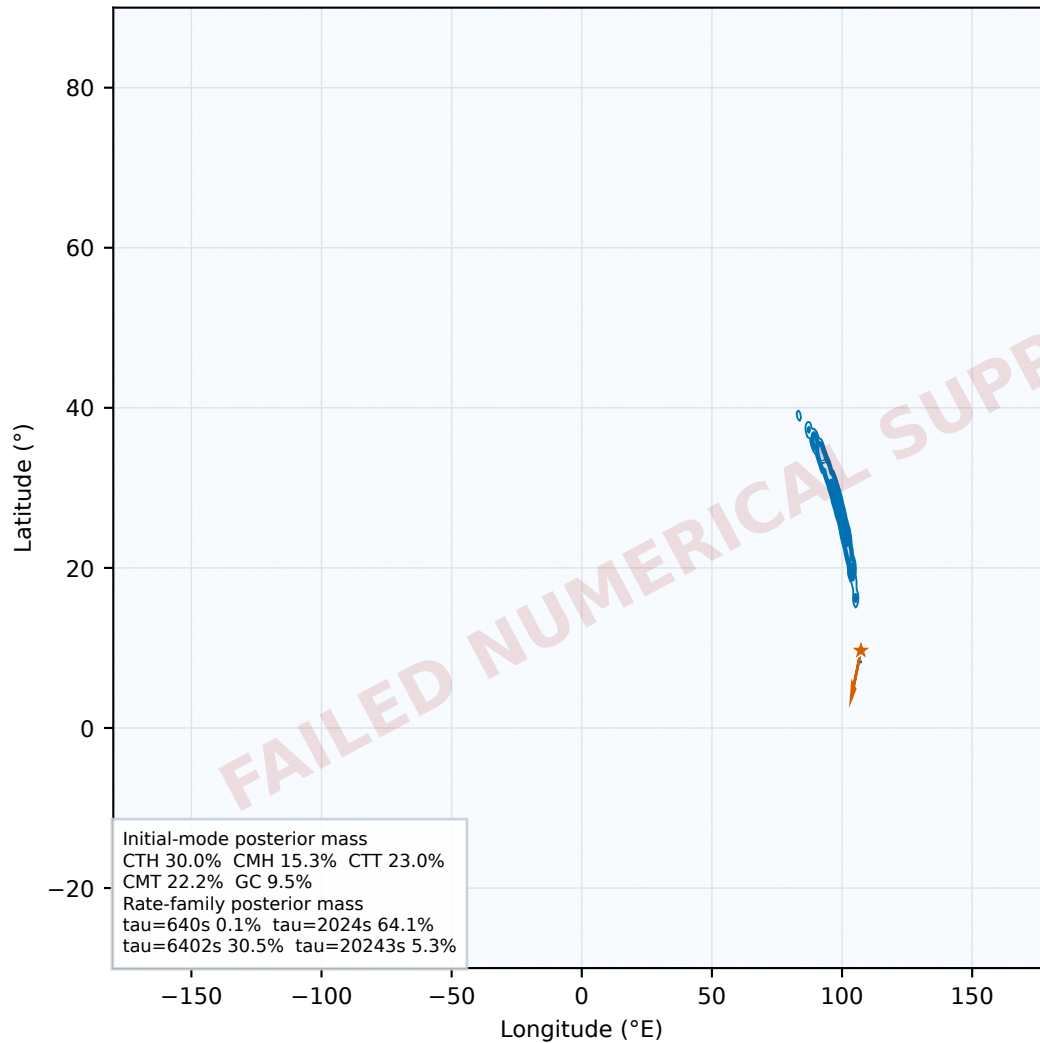


BTO + BFO (7 Hz) (kept separate)
SATCOM likelihood update: BTO + BFO
particle/root ESS 187 / 35.2
mean lateral/speed/altitude events 4.35 / 3.95 / 4.40
mass with ≥ 2 lateral events 95.9%
seed TV max 0.928
held-back mean error 504.3 NM
mass within 100 NM 3.4%

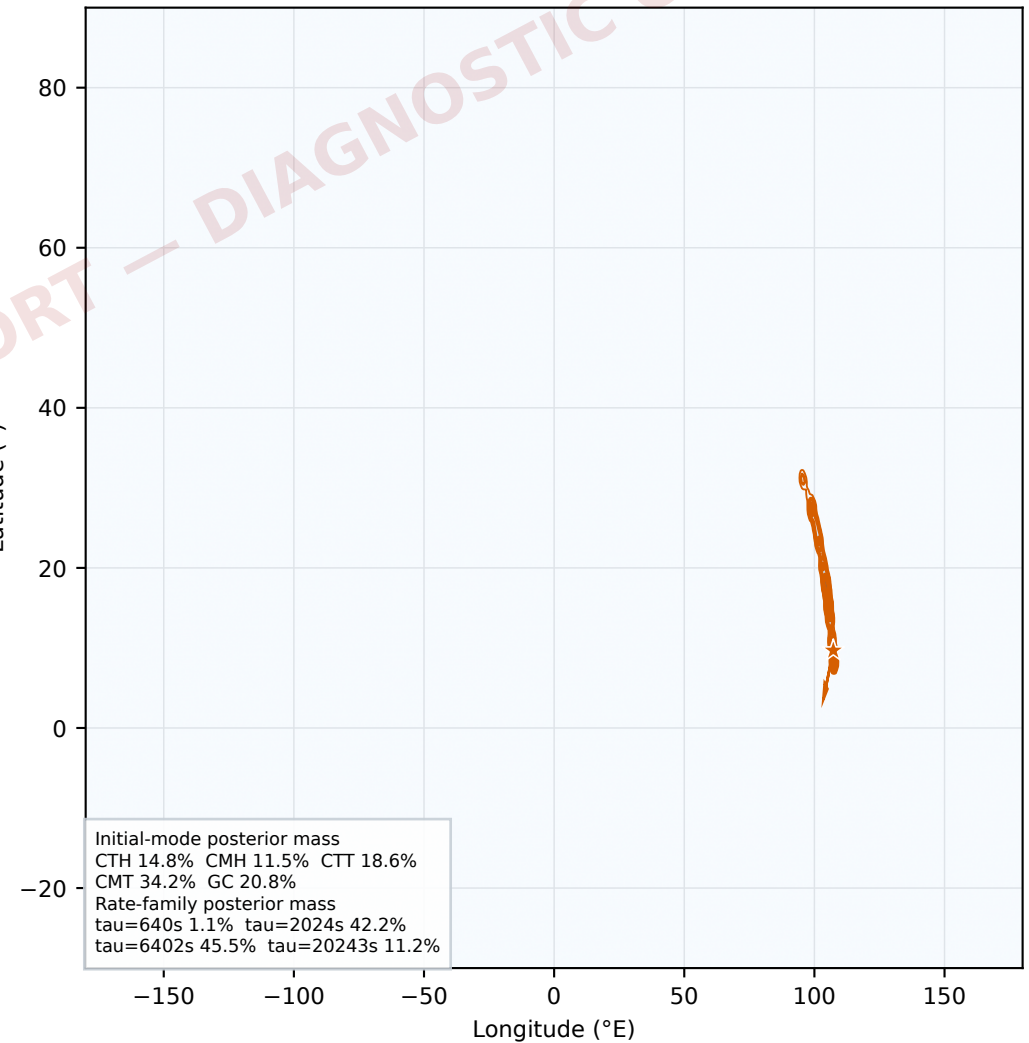


MH371 — mh371_0600 filtering posterior

BTO only (kept separate)
SATCOM likelihood update: BTO
particle/root ESS 196 / 30.2
mean lateral/speed/altitude events 5.04 / 5.57 / 5.97
mass with ≥ 2 lateral events 100.0%
seed TV max 0.994
held-back mean error 1176.8 NM
mass within 100 NM 0.1%

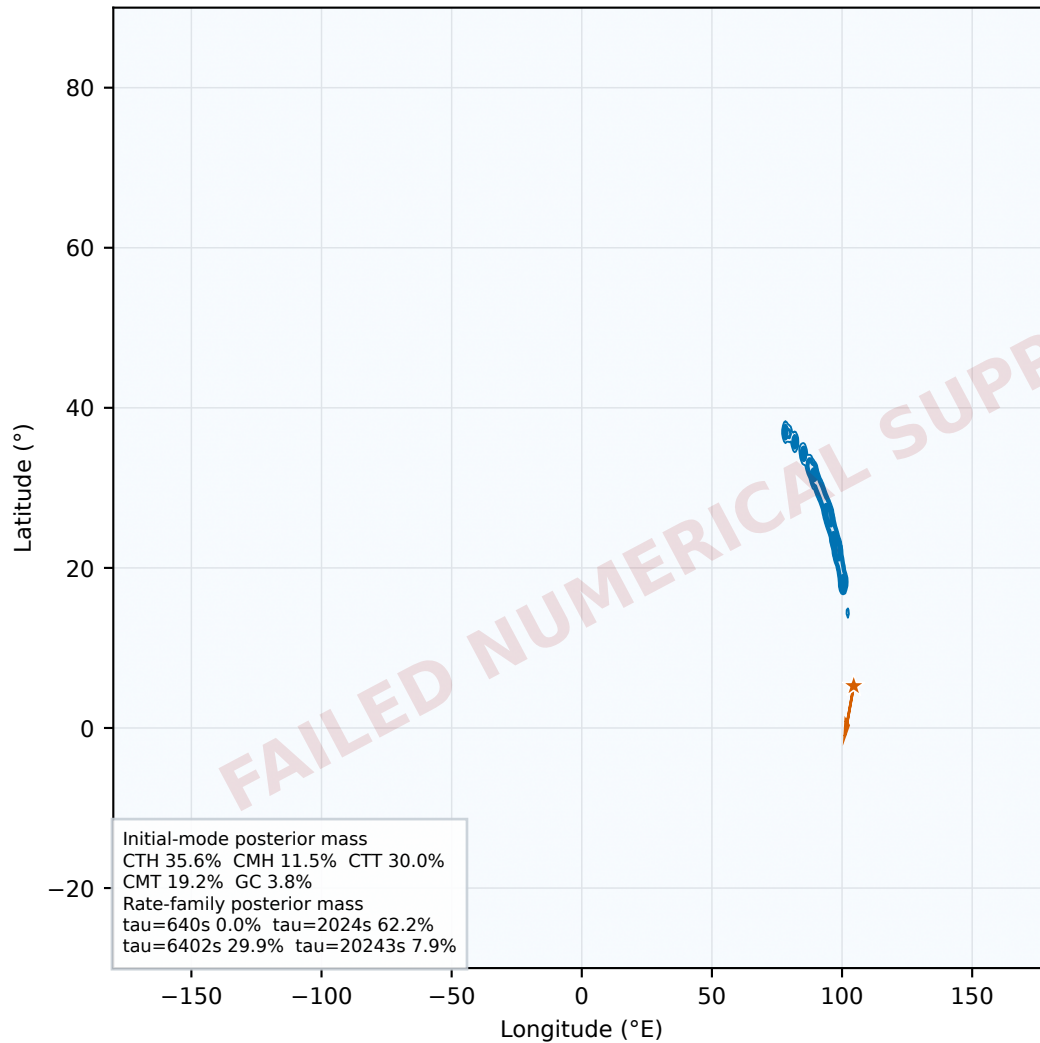


BTO + BFO (7 Hz) (kept separate)
SATCOM likelihood update: BTO + BFO
particle/root ESS 224 / 14.5
mean lateral/speed/altitude events 4.49 / 4.43 / 5.15
mass with ≥ 2 lateral events 100.0%
seed TV max 1.000
held-back mean error 513.5 NM
mass within 100 NM 20.5%

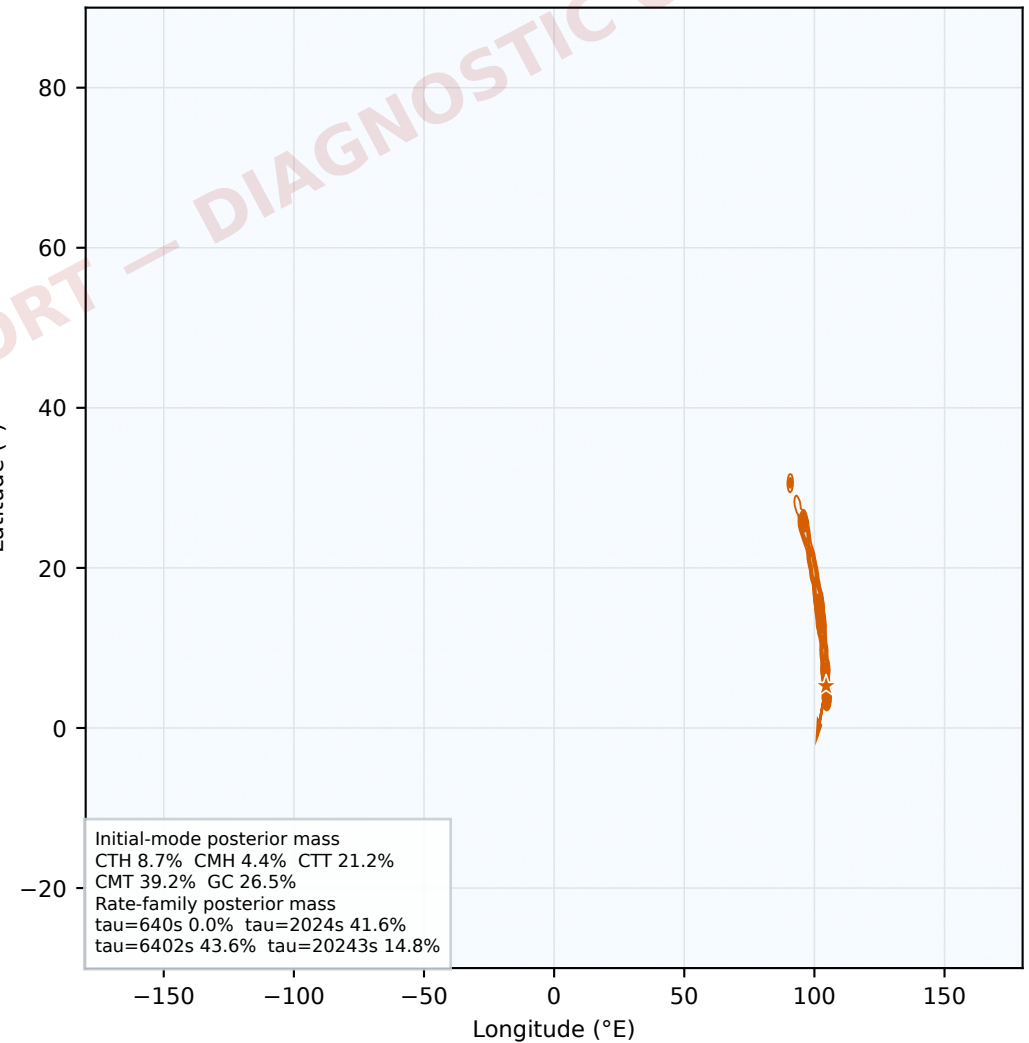


MH371 — mh371_0648 filtering posterior

BTO only (kept separate)
SATCOM likelihood update: BTO
particle/root ESS 711 / 22.4
mean lateral/speed/altitude events 5.38 / 6.44 / 6.59
mass with ≥ 2 lateral events 100.0%
seed TV max 0.973
held-back mean error 1350.4 NM
mass within 100 NM 0.1%



BTO + BFO (7 Hz) (kept separate)
SATCOM likelihood update: BTO + BFO
particle/root ESS 288 / 12.9
mean lateral/speed/altitude events 4.77 / 4.79 / 5.21
mass with ≥ 2 lateral events 100.0%
seed TV max 1.000
held-back mean error 559.9 NM
mass within 100 NM 14.2%



MH371 — structural sensitivity at mh371_0648

BTO only; groups remain explicit conditionals inside this declared family. Differences in conditional mean location are structural spread, not Monte Carlo error.

Five declared initial lateral modes

Declared group	Posterior mass	Conditional mean lat, lon	Particle / root ESS	Mean events L / S / A	Truth error NM
constant magnetic heading	11.50%	25.068, 95.780	203 / 8.7	6.83 / 6.39 / 4.69	1291.2
constant magnetic track	19.18%	23.413, 96.818	170 / 5.0	4.98 / 5.45 / 4.09	1177.4
constant true heading	35.59%	27.865, 92.946	181 / 6.7	5.50 / 6.09 / 7.47	1509.1
constant true track	29.95%	24.597, 96.217	215 / 5.4	4.91 / 7.23 / 8.00	1255.7
great-circle continuation	3.78%	29.800, 90.105	219 / 4.5	5.52 / 8.60 / 5.71	1684.3

Four declared recurring-event rate families

Declared group	Posterior mass	Conditional mean lat, lon	Particle / root ESS	Mean events L / S / A	Truth error NM
L=20243s/S=20243s/A=20243s	7.86%	18.485, 100.355	83 / 1.2	3.09 / 1.09 / 0.17	830.7
L=2024s/S=2024s/A=2024s	62.23%	26.460, 94.594	461 / 13.1	6.42 / 8.83 / 9.31	1394.1
L=6402s/S=6402s/A=6402s	29.89%	26.299, 94.081	182 / 9.0	3.81 / 2.85 / 2.61	1397.7
L=640s/S=640s/A=640s	0.01%	29.489, 91.830	5 / 1.1	21.08 / 24.92 / 30.93	1622.8

Masses are posterior within the named evidence family. Declared prior weights are equal and empirically uncalibrated; no cross-family mixture is formed.

MH371 — structural sensitivity at mh371_0648

BTO + BFO (7 Hz); groups remain explicit conditionals inside this declared family. Differences in conditional mean location are structural spread, not Monte Carlo error.

Five declared initial lateral modes

Declared group	Posterior mass	Conditional mean lat, lon	Particle / root ESS	Mean events L / S / A	Truth error NM
constant magnetic heading	4.43%	16.963, 100.018	22 / 4.4	6.09 / 4.66 / 4.83	750.8
constant magnetic track	39.23%	15.296, 100.761	163 / 4.1	3.22 / 4.04 / 4.18	641.9
constant true heading	8.65%	17.853, 100.413	31 / 2.7	7.21 / 9.00 / 8.30	793.6
constant true track	21.22%	12.699, 102.357	45 / 3.2	5.75 / 3.48 / 4.52	464.7
great-circle continuation	26.47%	11.848, 102.754	59 / 3.1	5.26 / 5.59 / 6.35	409.3

Four declared recurring-event rate families

Declared group	Posterior mass	Conditional mean lat, lon	Particle / root ESS	Mean events L / S / A	Truth error NM
L=20243s/S=20243s/A=20243s	14.77%	5.531, 104.333	206 / 1.4	3.00 / 0.20 / 0.86	19.2
L=2024s/S=2024s/A=2024s	41.58%	20.159, 98.979	120 / 6.5	5.92 / 9.83 / 9.88	951.2
L=6402s/S=6402s/A=6402s	43.65%	11.277, 102.990	99 / 5.4	4.28 / 1.54 / 2.24	372.6
L=640s/S=640s/A=640s	0.00%	27.331, 94.101	5 / 2.0	18.65 / 24.16 / 24.54	1452.6

Masses are posterior within the named evidence family. Declared prior weights are equal and empirically uncalibrated; no cross-family mixture is formed.